$$\dot{X} = F(X, t), \text{ where } X = \left[X_1, X_2, X_3, \dot{X}_1, \dot{X}_2, \dot{X}_3, \theta\right]^\top$$

$$\dot{X} = F(X, t), \text{ where } X = \left[X, Y, Z, \dot{X}, \dot{Y}, \dot{Z}, \theta\right]^\top$$

$$\dot{X} = F(X, t), \text{ where } X = \left[r, \theta, \phi, \dot{r}, \dot{\theta}, \dot{\phi}, \gamma\right]^\top$$

$$\dot{\underline{X}} = F(\underline{X}, t)$$

$$\underline{X} = \left[X, Y, Z, \dot{X}, \dot{Y}, \dot{Z}, \beta\right]^\top$$

$$\dot{X} = F(X, t)$$

$$X = \left[X_1, X_2, X_3, \dot{X}_1, \dot{X}_2, \dot{X}_3, \gamma\right]^\top$$

$$\dot{X}(t) = \dot{X}^*(t) + \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right] [X(t) - X^*(t)] + \cdots$$

$$x(t) := X(t) - X^*(t)$$

$$\dot{x}(t) = \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^* x(t)$$

$$A(t) := \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^*$$

$$A(t) := \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^*$$

$$\dot{x}(t) = Ax(t)$$